

CL-1000i Serise

Senario Maintenance Guide



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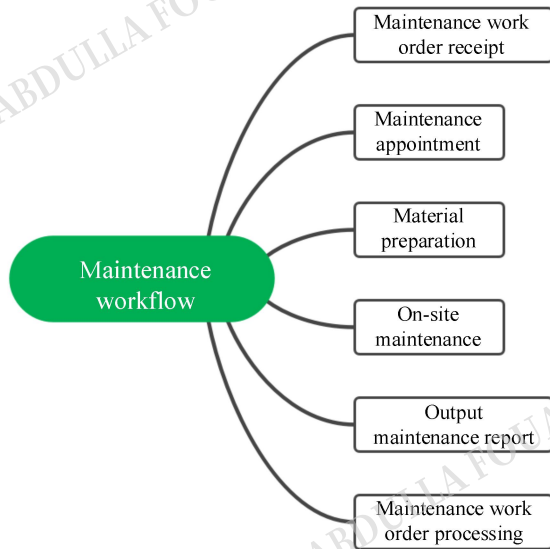
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Chapter 1 Preparation Before Maintenance & Appointment:

1.1 Maintenance Procedure:



- Maintenance should be carried out based on two criteria: Test times and use time. This applies to customers with in-warranty series products (including standard warranty and extended warranty).
- Since the last six-month maintenance, the maintenance will be carried out every 6 months or **36,000** tests.
- Since the last annual maintenance, the annual maintenance will be carried out every 12 months or **72,000** tests.
- Maintenance will be carried out after reaching either the test times or service time;
- Carry out on-site maintenance as required and actively demonstrate to customers the effect of maintenance and replacement of materials and maintenance;

1.2 Maintenance Item & Duration

Requirements:

Display the maintenance items by maintenance periods respectively;

No.	Category	Requirements	Maintenance Period	Maintenance Duration
1	Check waste connection and waste tank connection; check the status of consumables	The tubing shall be free of folding and damage, and the joint shall be free of leakage; confirm that all consumables are sufficient.	Six-month/One-year	2 min
2	Check the Software Version	Check the version of the operating software and refer to the latest version published on the TIS for upgrading	Six-month/One-year	1 min
3	Clean dust screens	Perform the visual inspection. Observe the dust on the dust screen in the light. No obvious blockage is acceptable. Restore it after drying thoroughly	Six-month/One-year	10 min
4	Check the Dispensing Syringe	No crystallization, liquid leakage or air bubbles are present. The lower end shall be against with the nut.	Six-month/One-year	3 min
5	Check the Dispensing Probe	The outer wall of the dispensing probe shall be free of dirt, and the inner wall of the probe shall be cleaned and inspected to ensure that the water outlet is smooth and linear	Six-month/One-year	5 min
6	Clean the Dispensing Probe Exterior	The outer wall of the dispensing probe shall be bright and free of dirt	Six-month/One-year	2 min
7	Clean the Wash Well	Visually inspect the wash well to ensure it is clean and that no overflow occurs during the dispensing probe cleaning process.	Six-month/One-year	2 min
8	Clean the Vortexer Hole	Clean the vortexer hole	Six-month/One-year	3 min
9	Clean the Lid of the Wash Buffer Tank	There is no crystallization on the surface of the tank lid cover, the exhaust hole of the tank lid is not blocked, and the sealing performance is good after installation.	Six-month/One-year	3 min

10	Clean the Gripper Finger	Fingers should be free of any dust, dirt, or rust and shows metallic luster	Six-month/One-year	2 min
11	Calibrate the Wash Buffer Bubble Sensor	The wash buffer bubble detection sensor is calibrated successfully	Six-month/One-year	3 min
12	Check the Overflowing of Dispersion Carousel	There is no wash buffer crystallization and solidified substrate attached to the lid of the dispersion carousel, dispersion chamber, and dispersion carousel, all of which shall show metallic luster.	Six-month/One-year	5 min
13	Replace/Clean the Dispersion Aspirating Probe	"Aspirate Flow Diagnosis" PASS in diagnosis	Six-month/One-year	10 min
14	Clean the Exterior of Dispersion Aspirating Probe	After cleaning, gently throw out any residual liquid from the tubing and wipe the surface of the probe.	Six-month/One-year	5 min
*	Replace the Dispersion Aspirating Peristaltic Pump Tubes	After replacement, there is no leakage in the tubing, and the dispersion aspirate flow is confirmed to be normal. Note: This maintenance item is only involved for older versions of the instrument (CL-1000i: Machines prior to YQ-6A000572 CL-1200i: Machines prior to YR-6A000051)	Six-month/One-year	15 min
15	Wipe the Dispensing Probe/Tube	No Crystallization; "System Check" shows Pass	Six-month/One-year	5 min
16	Align the Mechanical Position	Confirm the position of sampling probe/gripper/sample rack delivery/dispersion carousel unit	Six-month/One-year	30 min
17	Execute Daily Clean	Normal ending	Six-month/One-year	20 min
18	Run System Check	The diagnosis result is Pass	Six-month/One-year	40 min
19	Execute Effect Check	Normal ending	Six-month/One-year	15 min
20	Replace the Dispersion Aspirating Peristaltic Pump Tubes	Dispersion aspirate flow is normal	Annual	15 min
21	De-dust the Cooling Fans	Power on the whole unit to see if fans work properly. After installing back the panel, enter the software to query the status of the fans.	Annual	10 min
22	Replace the Spring of Gripper	Power on the instrument, and the gripper can normally grip the cuvettes.	Annual	10 min
23	Check the substrate leakage box	Visually inspect that substrate leakage box is clean and dry.	Annual	25 min
24	Check the Vacuum Waste Pump	The vacuum is established successfully	Annual	5 min
25	Check the Refrigeration Current	Check whether the current of the four Peltiers is between 2.0A and 6.5A	Annual	2 min
26	Statistics of Component Service Life	When a component is changed, select it on the Statistics of Component Service Life interface, and click Clear to erase its service life history.	Annual	2 min

Note:

- There are 26 proactive maintenance items for the CL-1000i. The following figure shows the recommended maintenance order.
- If some maintenance items do not belong to the maintenance period, skip them.
- Based on the table above, it can be roughly calculated that: Six-month maintenance takes approximately 170 minutes. Yearly maintenance takes the longest, at approximately 240 minutes. Please negotiate with the customer in advance to reserve sufficient maintenance time before performing proactive maintenance.

1.3 Maintenance Materials

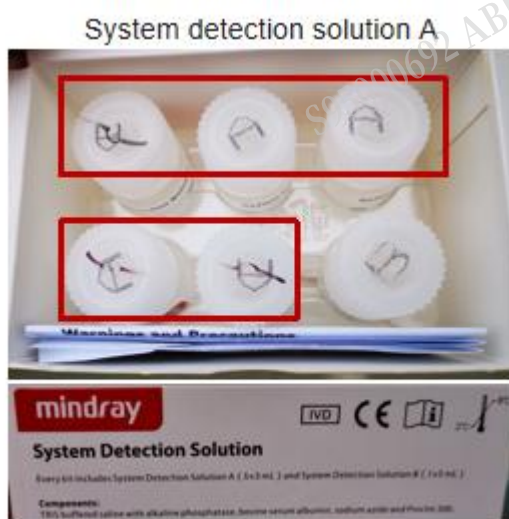
1.3.1 Maintenance Kit Required

After confirming the maintenance type, the engineer prepares materials in advance according to the form.

Some materials need to be applied for in the work order. For details, please refer to the following form: Note:

Acidic cleaning solution is only used when the substrate tubing is contaminated.

No.	Supplies and Tools	Applicable Maintenance Items	Remarks	Order number
1	Alignment tooling	Confirm and align the position	Engineer-provided	115-042852-00
2	Tool kit	Dismantle the analyzer components	Engineer-provided	/
3	CL-1000i&1200i Annual Maintenance Kit	Execute the six-month maintenance	SN: Before:YQ-6A000572/YR-6A000051,	115-035778-01
*	1-year maintenance kit	Execute the six-month maintenance	SN: After:YQ-6A000572/YR-6A000051 Start with W7- and BG-	115-040915-01
4	System Detection Solution A	For system tests	Order	105-009141-00
5	Acidic wash solution	For system tests	Order	105-009143-00
6	Calibrate the cuvette material package	Dispersion aspirate flow test	Maintenance work order request	115-089791-00



1.3.2 The maintenance Kit contains materials:

115-040915-01 1-year maintenance kit

No.	FRU Code	FRU Description	Quantity
1	0000-10-10916	acupuncture needle	2
2	033-000151-00	Finger clamping spring	2
3	082-000328-00	1ml Syringe	1
4	082-002512-00	Pump Tube, Double Segment ID 1.59 mm, Buckle Distance 90mm	8
5	095-002872-00	Dispersion unclogging device	2
6	115-024096-00	MS Aspirating Needle Cleaning Tool	1
7	115-042833-00	Short needle and tubing for aspirating	1
8	115-073958-01	Substrate spike assembly (BM80)	2



1.3.3 Tools and fixtures required for maintenance

The following supplies and tools will be used for maintenance of the system.

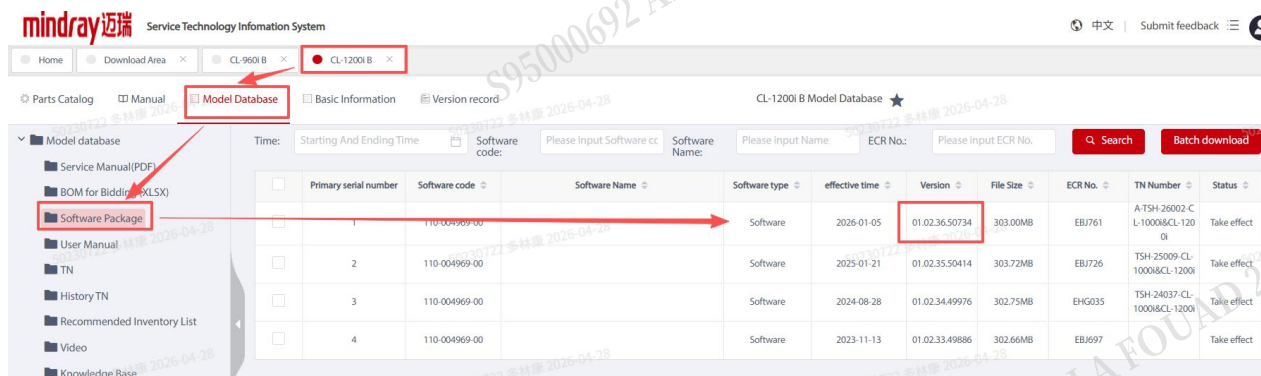
After confirming the maintenance type, the engineer shall confirm in advance that the materials are complete

according to the table. For details, please refer to the table below:

Materials and Tools Prepared by the Customer			
No.	Supplies and Tools	Applicable maintenance items	Provided by
1	Clean gauze	Clean the syringes and wipe the rotors, probes/mixers,dispersion dispensing probe exterior	Provided by the customer
2	Cotton swabs	Wash well maintenance, reagent compartment, clean mixer, etc.	Provided by the customer
3	Vacuum cleaner	Clean fans and dust screens, etc.	Provided by the customer
4	Brush	Clean dust screens	Provided by the customer
5	Thread syringe	Unclog the sample probes and reagent probes	Provided by the customer
6	Beaker	Probe unclogging and washing, etc.	Provided by the customer
7	Ethanol	Clean probes and dispersion mechanisms, etc.	Provided by the customer
8	Fiber-free gloves	Clean dispensing probes, dispersion aspirating probe exterior, etc.	Provided by the customer
9	CD80 Wash Solution	Clean the aspirating probes.	Provided by the customer
10	Hitachi cup	It is used for System Check	Provided by the customer

1.3.4 Software Required

- Latest operating software,TIS: After entering the URL <https://tis.mindray.com>, search by the following path



Chapter 2 During Maintenance

2.1 Software Operation

Manual Confirmation: After the maintenance personnel have completed the on-site manual maintenance, the maintenance procedure list displays all the manual maintenance procedures under the current maintenance period.

Select **Utility > Maintenance > Maintenance List/Engineer Maintenance** to manually confirm the maintenance items

The screenshot shows the 'Maintenance' software interface. At the top, there are tabs for 'Scheduled Maintenance' (labeled ①), 'Maintenance Command', and 'Engineer' (labeled ②). Below these are sub-tabs for frequency: 'Daily', 'Weekly', 'Two-week', 'Monthly', 'Three-Month', 'Six-Month', and 'Other'. The main area is a table with columns: 'Procedure', 'Select', 'Property', 'Operator', and 'Date Performed'. The table lists several procedures, each with a question mark icon and a checkbox. The 'Select' column has checkboxes for 'System' and 'User'. The 'Engineer' tab is selected, and the 'Check the Probe' and 'Clean Probe Exterior' rows have their 'System' checkboxes checked (labeled ③). At the bottom, there are buttons: 'Select All', 'OK' (labeled ④), 'Log', 'History', 'Customize', 'Delete', and 'Close'.

Procedure	Select	Property	Operator	Date Performed
Check Waste Tank Connection	☐	System		
Check Consumable Status	☐	System		
Check Syringe	☐	System		
Check the Probe	☒	System		
Clean Probe Exterior	☒	System		
Daily Clean	☐	System		
Effect Check	☐	System		

2.2 Maintenance Operation

2.2.1 Check the software version

Maintenance Object: Software Version

Maintenance Timing: Six-month

Maintenance Cause: Fix software bugs and improve ease of use

Maintenance Item: Check and update the software to latest version

Supplies Required: USB flash drive

Precaution: The USB flash drive used to copy the software package should be virus-free

Instrument Status: In the Idle or Stop status

Procedure:

- Enter the operation interface: Select **Utility > System Setup > Instrument > Version Info**; query the software version and determine whether the software version is the latest version; if not, download the

latest version software from TIS and upgrade the software to the latest version. Refer to the Software Upgrade section of the Service Manual for the software upgrade process.



The image shows a 'Software Version' dialog box with a blue header. It contains several input fields and buttons. The fields are: 'Software Version' with value '00.02.36.50734', 'Database Version' with value '242', 'Serial No.' with value 'YQ-5A000001', 'Model' with value 'CL-1000i', 'General parameters' with value 'V01.20', and 'Scanner Model' with value 'Honeywell 1950'. There are three buttons: 'Details' next to the Software Version field, 'Setup' next to the Scanner Model field, and 'Close' at the bottom right.

Field	Value
Software Version	00.02.36.50734
Database Version	242
Serial No.	YQ-5A000001
Model	CL-1000i
General parameters	V01.20
Scanner Model	Honeywell 1950

Confirmation method: Restart the instrument, select **Utility > System Setup > Instrument > Version Info** to check if the software is updated to the latest version.

Maintenance Duration: 30min

2.2.2 Check waste connection, waste tank connection, and the consumable status

Maintenance Object: Waste tube/waste tank; consumables

Maintenance Timing: Six-month

Maintenance Cause: During the use of the instrument, ensure that the waste discharge pipeline is smooth and there is no crystallization at the connector. If the waste discharge pipeline is blocked, the efficiency of the waste pump may be reduced, resulting in overflow of the wash well, insufficient vacuum and other faults.

Maintenance Item: Tubes and joints

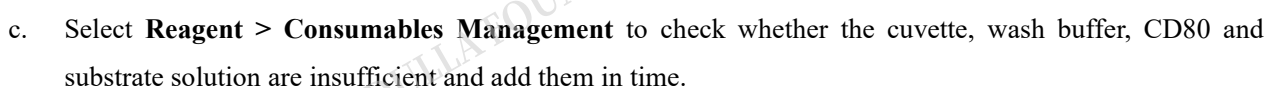
Supplies Required: None

Precautions: Wear gloves and lab coats before operation

Instrument Status: Standby or shutdown

Procedure:

- Check that the waste tubing runs smoothly without bending.
- No crystallization deposits at the waste liquid connector. If waste tank is used to hold the liquid waste, check the waste tank and empty it.



2.2.3 Clean Dust-proof Nets

Maintenance Timing: Six-month

Maintenance Cause: Dust may accumulate on the dust screens when the instrument is used for a long time, influencing the ventilation and heat elimination effects

Maintenance Item: Clean dust screen

Supplies Required: Vacuum cleaner, brush and fresh water

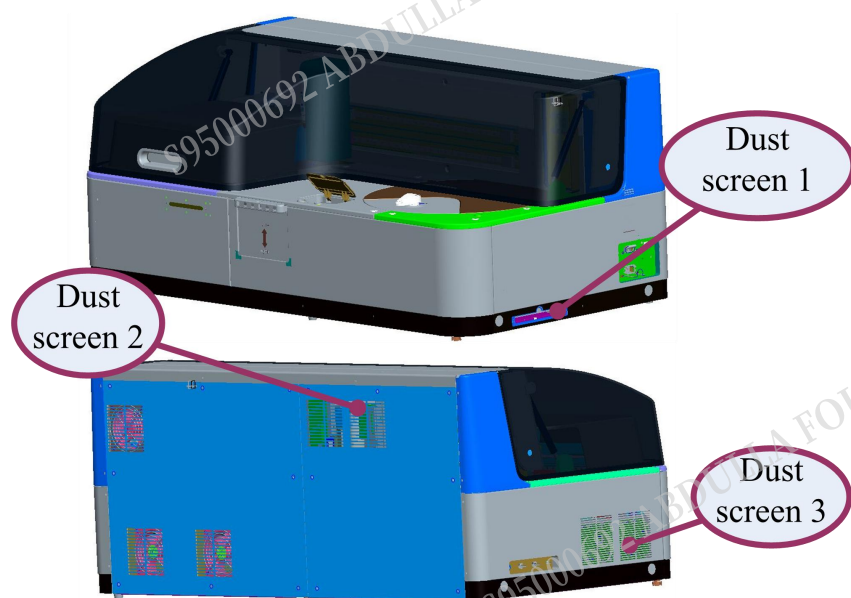
Precautions:

- a. Use a vacuum cleaner to clean the dust screens while keeping them installed.
- b. Use a brush and fresh water to clean the dust screens after removing them.
- c. Do not reinstall the dust screens until they are dry completely.
- d. Install the dust screens correctly to avoid gaps.
- e. There are 3 dust screens in the analyzing unit, all of which must be maintained.

Instrument Status: Any

Procedure:

- a. Verify that the instrument is powered off; open the front door of the instrument.
- b. Remove dust-proof net 1 from the right side of the instrument; open the left and rear back shield covers, and remove dust-proof nets 2 and 3
- c. For dust-proof net 1, you can hold it inward with both hands and lift it up, and then take it out. For dust-proof nets 2 and 3, you should remove the left and rear back shield covers of the instrument.



- d. Use the suction cleaner, or hair brush and fresh water to clean the dust-proof nets, and then dry them in air.
- e. Reinstall the dust-proof nets when they are dry.
- f. Close the front door of the analyzer.
- g. Power on the analyzer and run the operating software.
- h. Make sure that the system status is Incubation or Standby.
- i. Select Utility > Maintenance > Maintenance, and then select Scheduled Maintenance > Monthly.
- j. Tick the Select checkbox in the same row as Clean Dust-proof Nets.
- k. Click OK to refresh the current date as the performance date.
- l. Click Log and then record comments and other important information for the procedure.
- m. Click OK to save your input information.

Confirmation Method: Visually check the cleaning effect. Observe the dust on the dust screen in the light. No obvious blockage is acceptable.

Maintenance Duration: 10min

2.2.4 Clean cap of the wash buffer tank

Maintenance Object: Wash buffer tank lid

Maintenance Timing: Six-month

Maintenance Cause: When loading and replacing the wash buffer, the connector on the tank lid needs to be disconnected. At this time, liquid drops will fall on the tank lid. To prevent liquid drops from falling off on other components, or the air vent on the tank lid from being blocked by the dried crystals, this maintenance shall be carried out.

Maintenance Item: Clean the wash buffer quick plug and vent hole

Supplies Required: Clean gauze, DI water, TIP, and cotton swabs

Precautions: Keep spring return normal

Instrument status: Idle, Stop, or Downtown

Procedure:

- a. Pull out the wash buffer tray.
- b. Remove the cap and use clean tip head to gently strip the blockage.



- c. Wash the cap with DI water.
- d. Wipe it dry with clean tissue and reinstall the cap

Confirmation Method:

Please check again if the wash buffer tank lid is clean.

Maintenance Duration: 3min

2.2.5 Calibrate the Wash Buffer Bubble Sensor

Maintenance Object: Wash buffer bubble detection sensor

Maintenance Timing: Six-month

Maintenance Cause: After the instrument is used for a long time, the feedback signal of the bubble detection sensor may drift and need to be re-calibrated

Maintenance Item: Calibrate the wash buffer bubble detection sensor

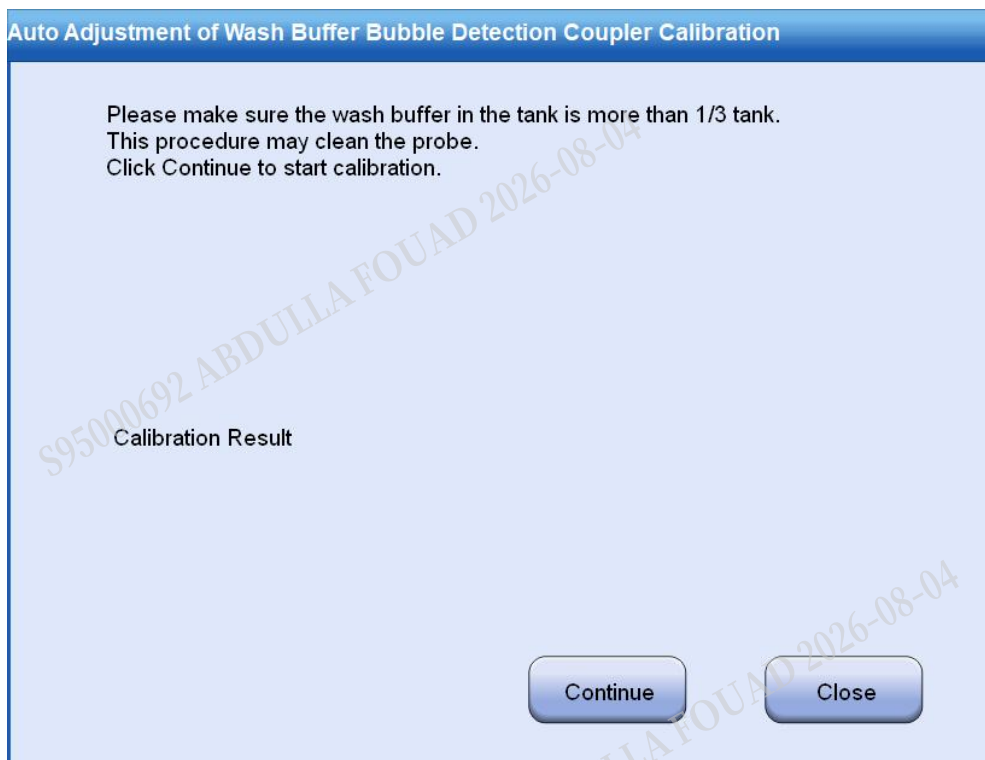
Supplies Required: None

Precautions: Wear gloves before operation to prevent biohazards

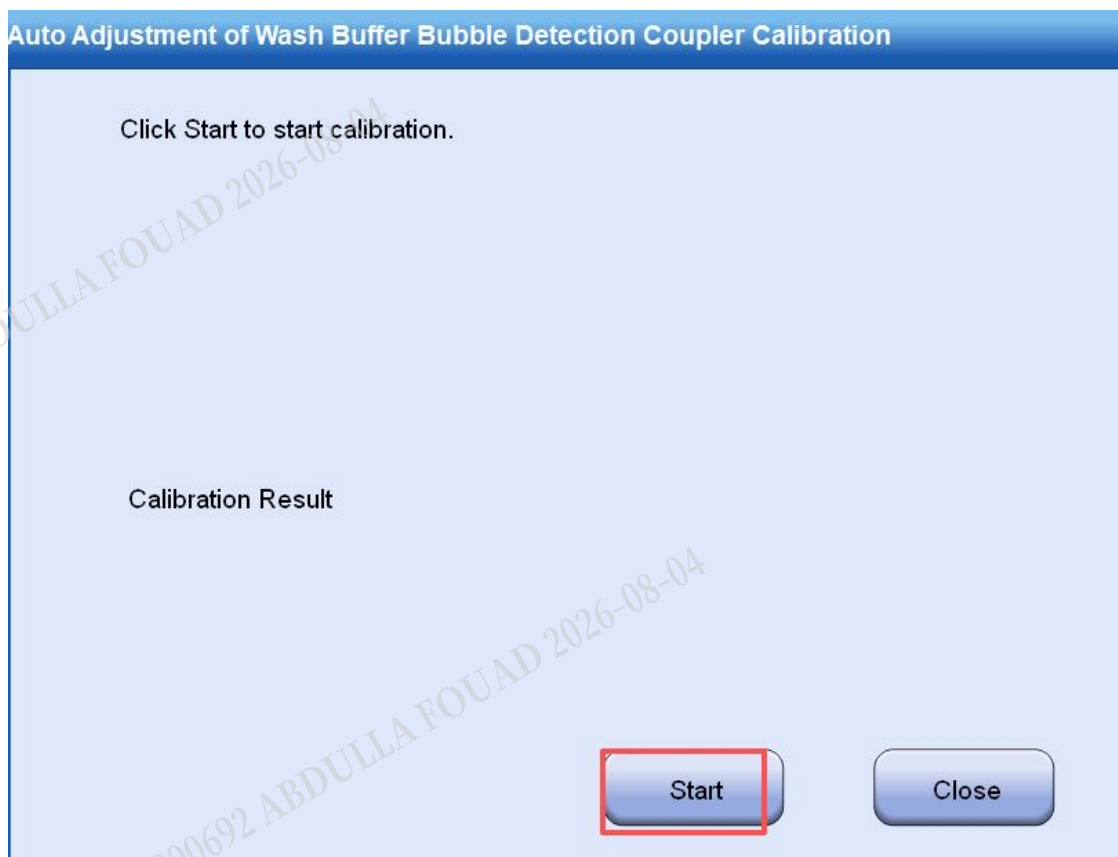
Instrument Status: Idle

Procedure:

- a. Select **Utility > Maintenance > Maintenance > Maintenance command**.
- b. Click **Sensor Calibration** command, and the wash buffer bubble sensor calibration interface will pop up.



- c. Make sure the wash buffer in the tank is more than 1/3 level and click **Continue**.
- d. Click **Start** to calibrate the wash buffer bubble detection sensor.



- e. The result will be displayed when calibration is successful.
- f. Click **Done** to exit the calibration procedure for the wash buffer bubble detection sensor.

Confirmation Method: Wash buffer bubble detection sensor calibration is successful

Maintenance Duration: 3min

2.2.6 Clean the Vortexer Hole

Maintenance Object: Two mixers

Maintenance Timing: Six-month

Maintenance Cause: The accumulation of dust in the vortexer hole will affect the mixing effect and contaminate the cuvette, which may cause problems related to the test results.

Maintenance Item: Clean the mixers

Supplies Required: Clean cotton swabs and ethanol

Precautions: Wear gloves before operation to prevent biohazards

Instrument Status: In the Idle or Stop status

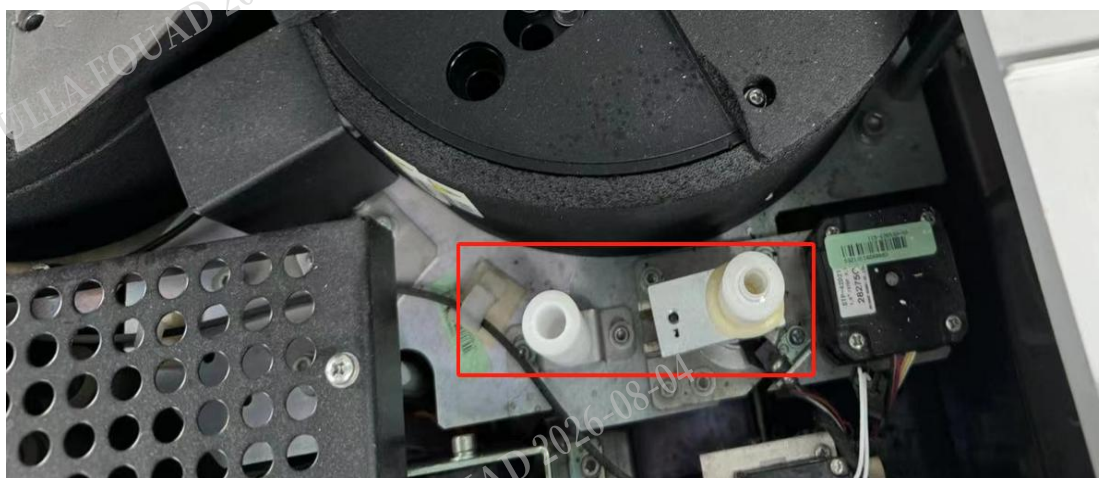
Procedure:

- a. Enter the operating software and then the Maintenance Command interface. Select **Clean Vortexer** and follow the software prompts to finish the procedure. At this point, the gripper motor is powered off. Move the gripper manually to create space for mixer maintenance.
- b. Wipe the vortexer hole gently with the cotton swab moistened with ethanol.

c. After the maintenance is completed, click **OK** to end the maintenance item.

Confirmation Method: Clean the mixers

Maintenance Duration: 3min



Note: After maintenance, please check whether there are used articles left on the maintenance parts;

2.2.7 Clean the Probe Exterior and Wash Well

Maintenance Object: Sampling probe/wash well

Maintenance Timing: Six-month

Maintenance Cause: If the sampling probe is dirty, it will cause carryover between samples or reagents and result in inaccurate results. In case of any crystallization in the wash well, the probe cleaning effect may be impaired, and the test results may be inaccurate

Maintenance Item: Clean the exterior of sample probe/reagent probe and the wash well of sample probe/reagent probe

Supplies Required: Clean gauze or cotton swab, ethanol, DI water

Precautions: Wear gloves and lab coats before operation

Pay attention not to twist or scratch the probe. If the probe is twisted or damaged, replace it immediately. Otherwise, the test results may be inaccurate

Instrument Status: Incubation or Idle

Procedure:

- Make sure that the system status is in Incubation or Idle status.
- Open the front cover of the instrument.
- Open the operating software, and then the Maintenance Command interface. Click **Clean Probe Exterior** and then click **Continue** on the pop-up window.
- After the dispensing probe returns to the wash well position, both the horizontal and vertical motors are de-energized. Manually move the dispensing probe arm to a position that allows easy access for wiping.

Using a gauze moistened with ethanol, gently wipe the probe exterior, and pay special attention to wipe the tip until the probe surface is clean and free of contaminants. Then, use a gauze moistened with DI water to remove any ethanol residue from the probe. Note: Do not push or pull the probe laterally, as this may bend it and affect normal test performance;

- e. Manually rotate the dispensing probe arm to a position easy for wash well maintenance. Use the dust-free cloth or cotton swab moistened with ethanol to gently clean the wash well until the wash well is free of any dirt. Wipe the ethanol off with the dust-free cloth or swabs moistened with DI water.
- f. Click **Done**; and the system starts to reset the probe.

Confirmation Method: Visually check if the sampling probe/wash well is clean.

Maintenance Duration: 4min



2.2.8 Check the Sample probe Syringe

Maintenance Object: Sample probe syringe

Maintenance Timing: Six-month

Maintenance Cause: The syringe is precise devices used to aspirate/dispense small amount of sample and reagent. If the syringe leaks or the nut under the syringe is not tightened, it cannot aspirate/dispense the correct amount of sample or reagent, and may even be damaged. Prior to measurements every day, check the syringe for leak.

Maintenance Item: Check the sample/reagent syringe for leakage and air bubbles.

Supplies Required: Dry gauze or cotton swabs

Precautions: Wear gloves before operation to prevent biohazards

Instrument status: Idle, Stop, or Downtown

Procedure:

- a. Open the shielding cover and you can see the syringe on the rear right of the analyzer.



b. Check the syringe T-piece assembly for leakage. Use the dry gauze or dry tissue to wipe the T-piece assembly, and then check if the gauze is moistened. Check the syringe interior for air bubbles.

If there is leakage, tighten the T-piece assembly, and check it again. If the leakage remains, check if the washer inside the syringe connector is intact. If the washer is damaged, replace it with a new one; otherwise, replace the syringe.

If there are air bubbles, take actions according to the syringe air bubble removing procedure.

Attachment: Procedure for bubble removal :

1. Select Utility > Maintenance > Maintenance > Maintenance Command;
2. Click Syringe Maintenance and click Continue,
3. Loosen the screws on the syringe shield at the right back side of the analyzer unit, and then remove the shield to see the syringe;(Step 3 is only relevant for early instrument)
4. Loosen the four remaining screws on top of the syringe counterclockwise, and then remove the screws and the fixing blocks.
5. Loosen the fastening screws at the bottom of the syringe counterclockwise and then remove the syringe from the frame;
6. Hold the Tpiece with one hand and the syringe connector with the other hand. Loosen the syringe counterclockwise and then remove the washer.
7. Soak the new syringe connector in the DI water beaker, pull the plunger head to aspirate half syringe of DI water, and then push the plunger head to remove the air. Repeat this operation to remove all air bubbles. Aspirate half syringe of DI water to prevent new bubbles.
8. Put the washer in the Tpiece. Hold the Tpiece with one hand and the syringe connector with the other hand, and then screw the Tpiece clockwise.
9. Install the syringe on the bracket.
10. Install the fixing blocks and the 4 fastening screws, but do not tighten the screws.
11. Align the plunger head to the fastening screws at the bottom of the syringe, and then tighten screws clockwise.
12. Pinch the plunger guide cap to adjust the syringe height. For the sample syringe, make the syringe head over the upper fixing block for 15 scales.
13. Tighten the four fastening screws on the fixing blocks.
14. After finishing the replacement, click Continue. The system resets the syringe unit. Check if the new syringe has leak. If it does, perform the Check Syringes procedure to check the syringe.
15. Select Fluidic Priming to execute the probe cleaning and priming for 15 times;
16. Select Utility > Maintenance > Maintenance > Scheduled Maintenance > Other.

17. Mark the checkbox corresponding to the Remove Air Bubbles from Dispensing Syringe.
18. Click OK. The system will refresh the maintenance time of each maintenance item.
19. Click Log, which records errors that occurred and other necessary information for the procedure.
20. Click OK to save the log.

- c. Check the plunger guide nut at the bottom of the syringe for leakage. Use the dry gauze or dry tissue to wipe the T-piece, and then check if the gauze is moistened.

If there is leakage, tighten the plunger guide nut, and check it again. If the leakage remains, replace the syringe.

- d. Check if the retaining nut at the bottom of the syringe is tightened. If not, tighten it with your hand (be careful not to strip the threads or cause breakage). If yes, proceed to the next step.

Confirmation Method: Observe whether the syringe leaks. Wipe the syringe connector assembly and plunger with the dry gauze or dry tissue to see if there is any leakage.

Maintenance Duration: 3min

2.2.9 Check the dispensing probe

Maintenance Object: Sampling probe

Maintenance Timing: Six-month

Maintenance Cause: Abnormal probe may affects the test results. Similarly, if the wash well does not dispense water properly, the probe cannot be effectively cleaned, which may affect test results. Prior to measurements every day, check the probe for stains and crystals, and check if the wash well is clogged. If the above-mentioned abnormalities exist, clean or adjust the probe immediately.

Maintenance Item: Check the dispensing probe for water dripping, stains and liquid flow abnormalities. Check the wash well for normal water dispensing.

Supplies Required: None

Precautions: Wear gloves and lab coats before operation

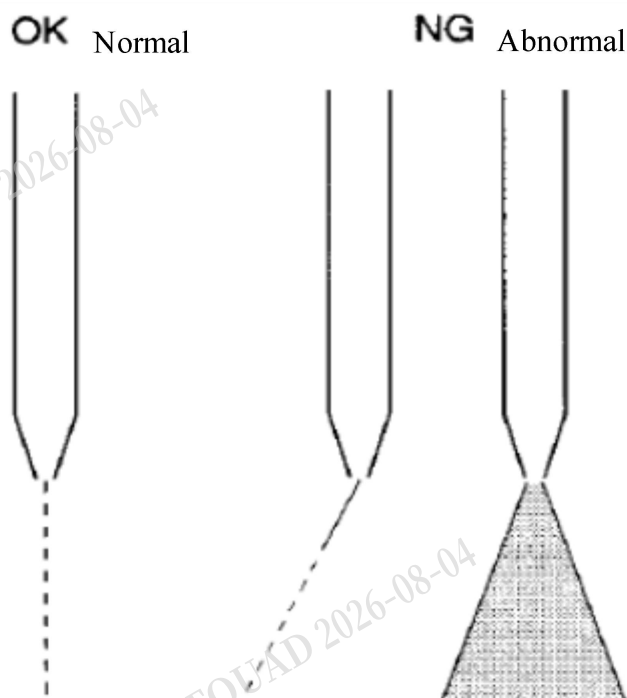
Pay attention not to twist or scratch the probe. If the probe is twisted or damaged, replace it immediately. Otherwise, the test results may be inaccurate

Instrument Status: Incubation or idle status

Procedure:

Make sure that the system status is in Incubation or Idle status.

- a. Open the front cover of the instrument.
- b. Open the operating software, select **Clean Probe Interior** and select **Continue** in the Maintenance Command interface. The probe moves to above the wash well and is washed for 3 three times;
- c. Observe the liquid flow of the probe interior as shown in the figure below;



Probe interior wash liquid flow

If the wash buffer sprays or does not come out vertically from the tip, the probe may be clogged.

Solution: First, perform the probe unclogging and **Daily Clean** maintenance. If the issue persists, proceed with **Clean the Dispensing Probe Interior**. If the problem still persists, consider replacing the dispensing probe.

- d. After the **Clean Probe Interior** is completed, the system starts to reset.

Confirmation Method: Repeat the above procedure to confirm the maintenance results.

Maintenance Duration: 5min

2.2.10 Check the Overflowing of Dispersion Carousel

Maintenance Object: Dispersion carousel

Maintenance Timing: Six-month

Maintenance Cause: Due to factors such as dispersion probe blockage or mis-operation, the dispersion carousel is prone to overflow. Therefore, it must be inspected and cleaned regularly. Failure to do so may result in ineffective dispersion mixing and increase the risk of abnormal test results.

Maintenance Item: Check the dispersion carousel for overflow. If so, clean the carousel.

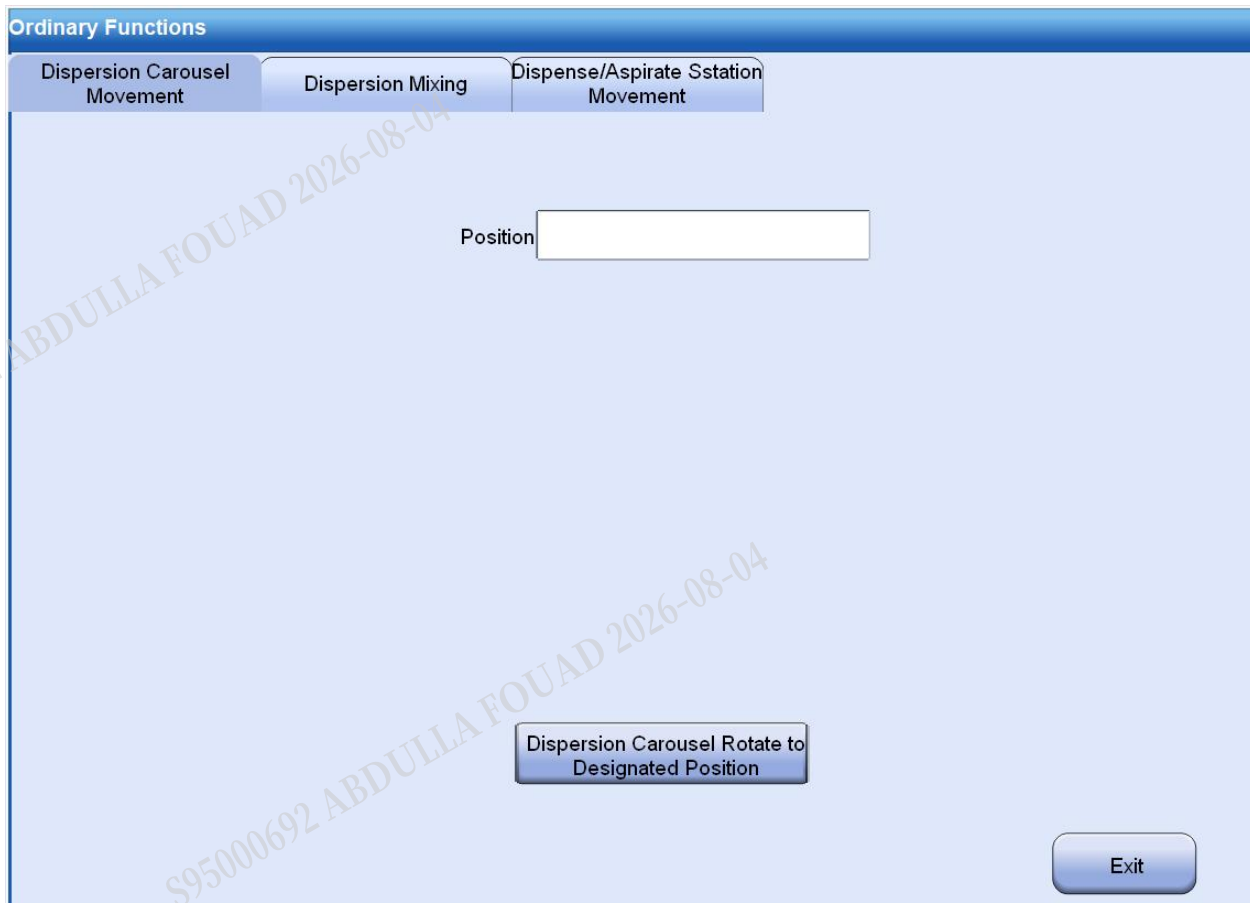
Supplies Required: Swab, dust-free gauze, and ethanol

Precautions: Wear gloves before operation to prevent biohazards

Instrument Status: Startup or Idle

Procedure:

- a. Select **Utility > Maintenance > Alignment > Dispersion Carousel > Ordinary Functions**.



- b. When a single cuvette is delivered to the dispersion carousel, from the direction designated by the red arrow in the following figure, observe the 28 cuvette positions of the dispersion carousel to see if there is wash buffer crystallization and yellow solidified substrate. If not, no overflow occurs. Click **Exit** to return to the previous interface, and then click **Dispersion Reset** to perform a mechanical reset of the dispersion carousel.



- c. If overflow occurs, dismantle the dispersion carousel and clean it.
For detailed steps, refer to Replacing Dispersion Carousel Heater of the Service Manual.
- d. **Cleaning Method:**
- 1) Flush the dispersion carousel with water and dry it with clean cloth.
 - 2) Use dust-free gauze moistened with ethanol to clean the interior of the dispersion chamber. Dry it with

dust-free gauze.

- 3) Use the dust-free gauze moistened with ethanol to clean the wash buffer crystallization or solidified substrate off the lower surface of the dispersion carousel lid and the gripping hole, then dry them with dust-free gauze.
- 4) To clean the substrate dispersing probe after overflow: Use the gauze moistened with ethanol to gently wipe the exterior of the substrate spikes, then use the gauze moistened with DI water to wipe off the ethanol. Note: Be careful not to damage the substrate tubes.
- 5) To check the resilience of the cuvette presser and the substrate mixing presser: Press the pressers with your gloved finger. If the pressers can slide smoothly in the mixing shaft without jamming, the resilience of the pressers is good. Replace the reaction cuvette presser or substrate mixing presser assembly in case of jamming.
- 6) To check the reliability of dispersion bearing after overflow: Rotate the 28 holders on the dispersion carousel assembly with your gloved finger. If each holder rotates smoothly without jamming, the dispersion bearing is confirmed to be reliable. If jamming occurs, replace the dispersion carousel bearing.
- 7) Clean the gripper according to the gripper maintenance guide.
- 8) After the overflow, check whether there is crystallization in each hole of the reaction disk; if so, the reaction carousel needs to be cleaned.
- 9) After the above procedures are complete, restore the dispersion carousel assembly.

Confirmation Method:

After the cleaning process in the step above, guarantee no crystallization and solidified substrate attached to the dispersion carousel lid, dispersion chamber, and dispersion carousel, all of which shall show metallic luster.

Maintenance Duration: 5 min (without overflow)/60 min (with overflow)

2.2.11 Dispersion Aspirating/Dispensing Probe Maintenance

Maintenance Object: Dispersion aspirating probe (4 pcs), and dispersion dispensing probe (4 pcs)

Maintenance Timing: Six-month

Maintenance Cause:

- The inner wall of the dispersion aspirating probe may be dirty after being used for a period of time, resulting in poor cleaning effect of dispersion.
- The wash buffer may crystallize on the outer wall of the dispersion aspirating probe after a long time of use. If the wash buffer falls into the dispersion carousel, the cleaning effect of dispersion may become poor.

Maintenance Item:

- Replace the aspirating probes with the clean ones.
- Clean the replaced probes for spare use.

Supplies Required: 4 clean aspirating probes, unclogging tools (including 1 rod unclogging tool and 1 syringe unclogging tool), clean non-crumb tissues, deionized water, wash buffer, dispersion aspirating probes, and wash syringe

Precautions:

- Wear gloves before operation to prevent biohazards
- To avoid being injured by the probes, do not bend the aspirating probe.

Instrument Status: In the Idle or Stop status

Procedure:

- a. Select **Utility > Maintenance > Maintenance Command > Clean Dispersion Probes/Tubes.**



- b. Select **Continue**; the dispersion carousel is mechanically reset, and the aspirating plate moves to a position convenient for maintenance.

c. To complete the dispersion dispensing probe maintenance:

- 1) Unscrew counterclockwise to loosen the round nuts on the dispensing probes and remove the dispensing probes (D1-D4) from the dispersion carousel.
 - 2) Use the clean gauze moistened with ethanol to wipe the exterior of the dispensing probes until they show metallic luster.
 - 3) Install dispensing probes back to the dispensing board and tighten the screws.
 - 4) Check if the tubing is loose. If so, tighten it.
- d. Complete the dispersion dispensing probe maintenance according to prompts:
- 1) Unscrew the round nut on the aspirating board counterclockwise, pull the 4 aspirating probes (A1-A4) upwards from the dispersion carousel. Pay attention not to bend or bump the probes.



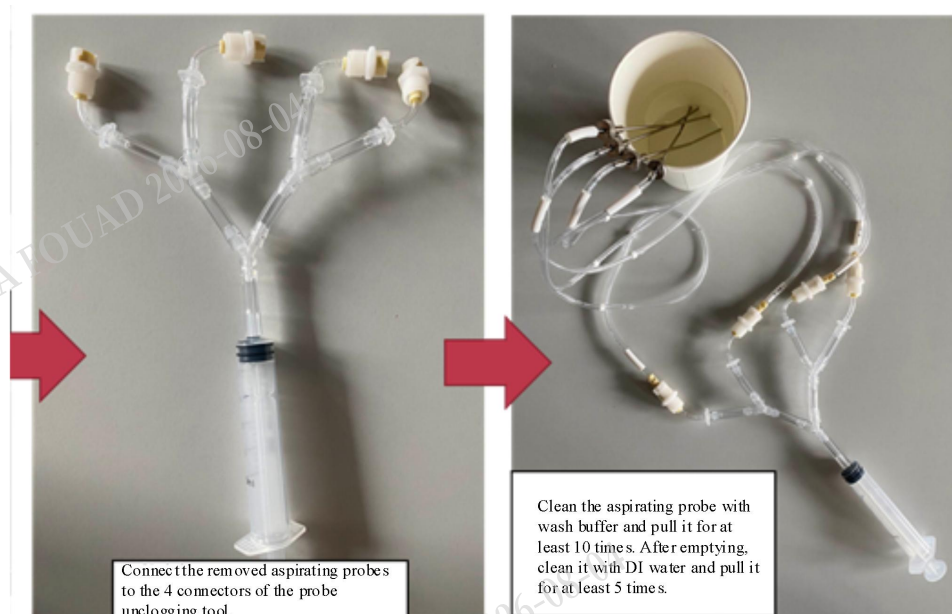
- 2) Connect one end of the aspirating tube to the probe barb, and the other end to the multi-way connector with the quick-connector. Disconnect the tube and the dispersion multi-way connector. **Do not attempt to disconnect the tube from the aspirating probe.**



- 3) Take a clean aspirating probe A4 and put the probe back into the aspirating board. During the process, take care to prevent the probe from scraping or colliding with other accessories. Turn the round nut clockwise to lock the aspirating probe. Gently lift the aspirating probe to check whether the aspirating probe is smoothly rebounded automatically. Connect the quick connector of the other guide tube of the aspirating probe with the quick connector on the dispersion aspirating adapter board. When connecting them, press the connectors gently and screw them clockwise until the buckles of the joints are fastened and a click is heard. Check whether the connector is fastened by shaking it gently. Fix the tube onto the press plate and the black ring symbol on the tube should align with the edge of the press plate.
- 4) Repeat steps to replace other 3 dispersion aspirating probes.

e. Wash the replaced aspirating probes:

- 1) Use the unclogging rod to clean and unclog the replaced aspirating probe.
- 2) After unclogging, connect a special syringe (supporting quick-connector connection) to the tube's quick connector. Prepare the probe wash buffer or CD80 wash solution. Immerse the other end of the probe into the wash buffer, and continuously pull and push the plunger of the syringe to clear the internal dirt of the probe.



- 3) Discharge the liquid from the aspirating probe, prepare no less than 100 ml of fresh DI water, immerse the probe tip in the liquid, and then pull and push the syringe plunger for no less than 10 times to clean its interior.
- 4) When the cleaning is completed, take the probe out of the DI water, and then pull and push the syringe plunger for five times to drain the liquid inside.
- 5) Remove the probe from the syringe, shake the liquid off it, and use clean gauze (without scraps) to dry the exterior of the probe.
- 6) Place the probe in a clean and ventilated environment to **dry it**.
- 7) When the probe is completely dry, keep it in a clean sealed bag for future use.
- f. Click **Done** and the dispersion unit resets the probe automatically.
- g. After replacement, select **Maintenance Command > Fluidic Priming**, and select **Dispersion Priming for 5 Times**, and then click **Execute** button. Observe the dispersion aspirating probe during the priming and ensure that the probe has been replaced correctly without tube leakage.

Confirmation Method: Aspirate Flow Diagnosis function shows PASS in diagnosis, and System Check shows Pass (it can be done later),

Maintenance Duration: 20min

2.2.12 Replace the Dispersion Aspirating Peristaltic Pump Tubes

Maintenance Object: Dispersion aspirating peristaltic pump tubes

Timing: One-year (Six-month/one-year for old models)

Maintenance Cause: Peristaltic pump is used for aspirate the dispersion wash buffer, of which the pump tubes may be worn out after long-term use.

To ensure the cleaning effect, the pump tubes should be replaced regularly

Maintenance Item: Replace the worn pump tubes to ensure the cleaning effect

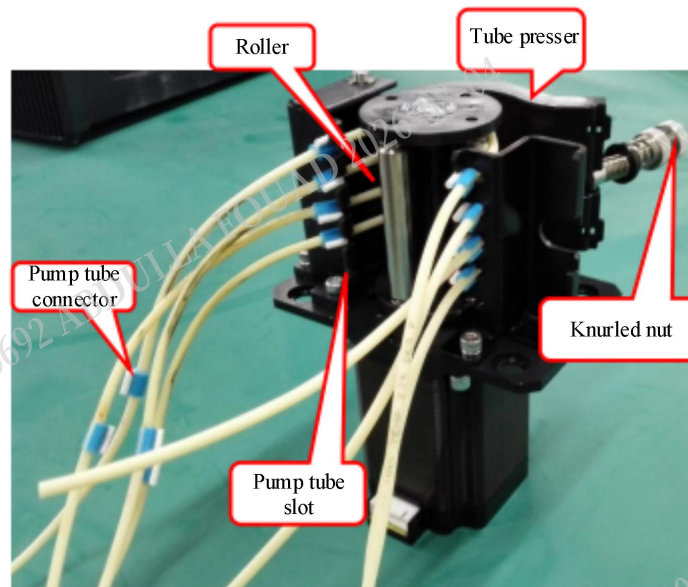
Supplies Required: A set of dispersion aspirating peristaltic pump tubes (4 pcs).

Precautions: Wear gloves before operation to prevent biohazards

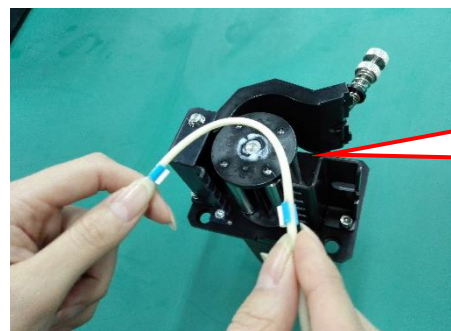
Instrument Status: The instrument is powered off.

Procedure:

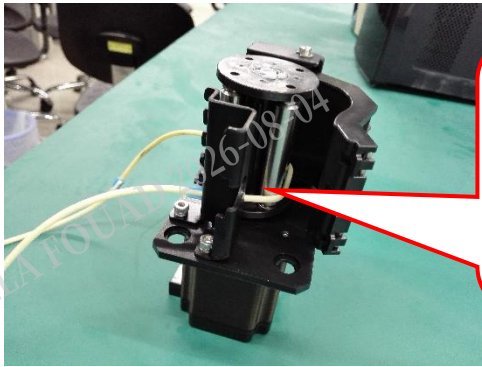
- a. Power off the analyzing unit, open the left rear shield cover of the instrument, and find the dispersion aspirate peristaltic pump.
- b. Use a tissue near the pump head to prevent liquid from spilling to other devices during the replacement process.
- c. As shown in the figure below: on the pump head, from top to bottom, there are 4 tubes in sequence. The bottommost slot is vacant, and no tube should be installed.



- d. When removing the tubes, first loosen two knurled nuts in turn, open the fixing blocks of tubes backwards, and take out the four tubes in the order from top to bottom. Be sure to protect the peristaltic pump tube during the disassembly process to prevent scratches.
- e. When replacing the tubes, leave the bottommost slot empty, and install the tubes from bottom to top. Each time you install a pump tube, you should turn the peristaltic pump roller by hand at least 10 times to ensure that each pump tube is in its own channel, and avoid cross-channel or intertwining. Use the same method to complete the replacement of the other three tubes. Note that the direction of the tubes must be correct, otherwise the dispersion washing will fail. If the direction cannot be confirmed, refer to other tubes that are not disassembled.



**When installed,
the pump pipe is
in an arc shape.**



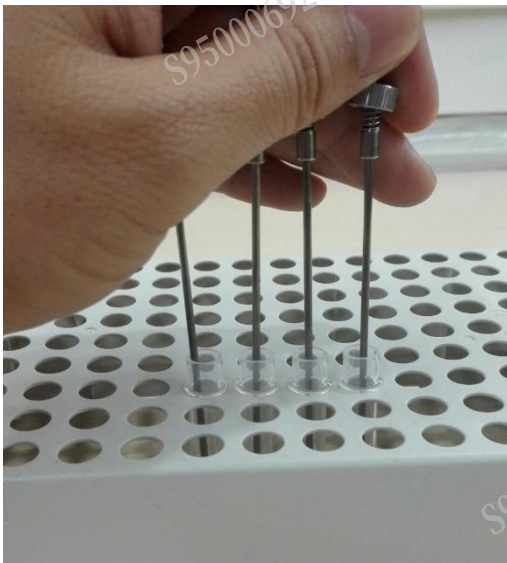
Each time you install a pump pipe, you should turn the peristaltic pump roller by hand at least 10 times to ensure that each pump pipe is in its own channel. Prevent the pump pipe from cross-channel or intertwining.

- f. After pump tube installation, fix the tube fixing blocks by tightening first the inner knurled nut and then the outer one. Do not tighten them simultaneously.
- g. Power on the complete machine and enter the next step for confirmation:

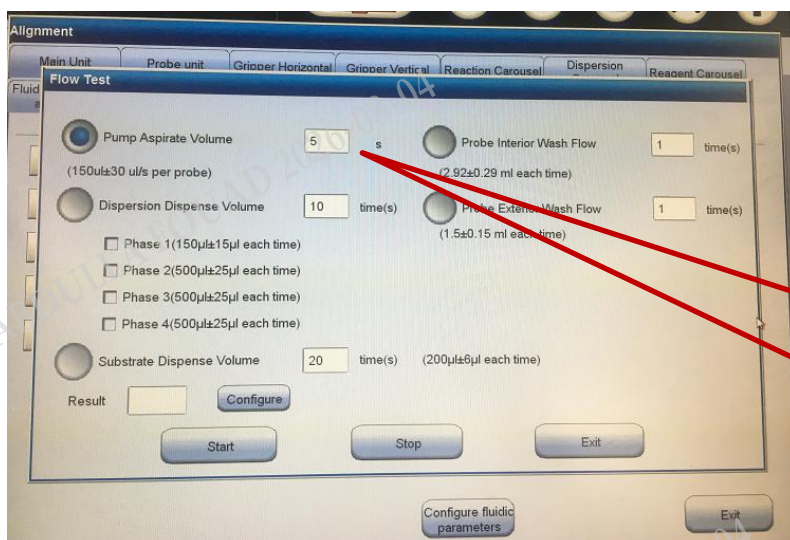
Dispersion aspirate flow test

Procedure:

- a. Prepare 4 empty cuvettes and mark them with A1, A2, A3, and A4 respectively. Use the calibrated pipette to add 1 ml of pure water into each empty cuvette, and place the four cuvettes on an empty cuvette tray.
- b. Remove 4 aspirating probes A1-A4 and put them into four cuvettes with water, respectively. Make sure the probe tip reaches the bottom of the cuvette and coincides with the axis of the cuvette without tilting.



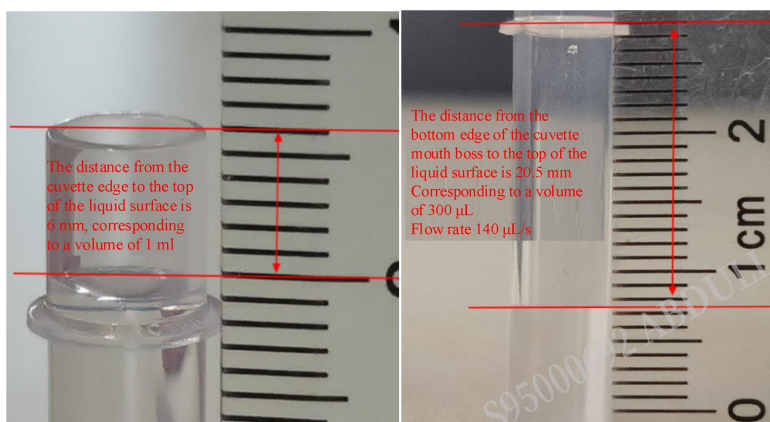
- c. Select Utility > Maintenance > Alignment > Fluidic Check and Calibration > Flow Test
- d. Select aspirate volume of the peristaltic pump, change the aspirating time from 7s to 5s, and click Start. After aspirating for 5s, the aspirating probe will aspirate part of the water in the cuvette:



Adopt the alignment index of $140 \pm 5 \mu\text{L/s}$, but ignore the interface index

- After the peristaltic pump stops, take out the aspirating probes A1–A4, use the 200 μL pipette to measure the liquid volume in the marked cuvettes A1–A4 and record the volume.
- Calculate the aspirate flow of the A1–A4 aspirating probes respectively.
- Aspirate flow = $(1,000 \mu\text{L} - \text{remaining liquid volume in the cuvette after aspiration} (\mu\text{L}))/5\text{s}$

Note: If the pipette cannot measure the volume accurately, a ruler can be used



Note: If a cuvette with scales is applied for and used, take out the A1-A4 aspiration probes after step 4 above and read the scale directly.

Confirmation Method:

Pass the dispersion aspirate flow test

Maintenance Duration: 15min

2.2.13 Clean the gripper finger

Maintenance Object: Gripper fingers

Maintenance Timing: Six-month

Maintenance Cause: The gripper may be polluted after working for a long time, and shall be cleaned. Otherwise, gripping failure may be caused.

Maintenance Item: Clean the gripper fingers

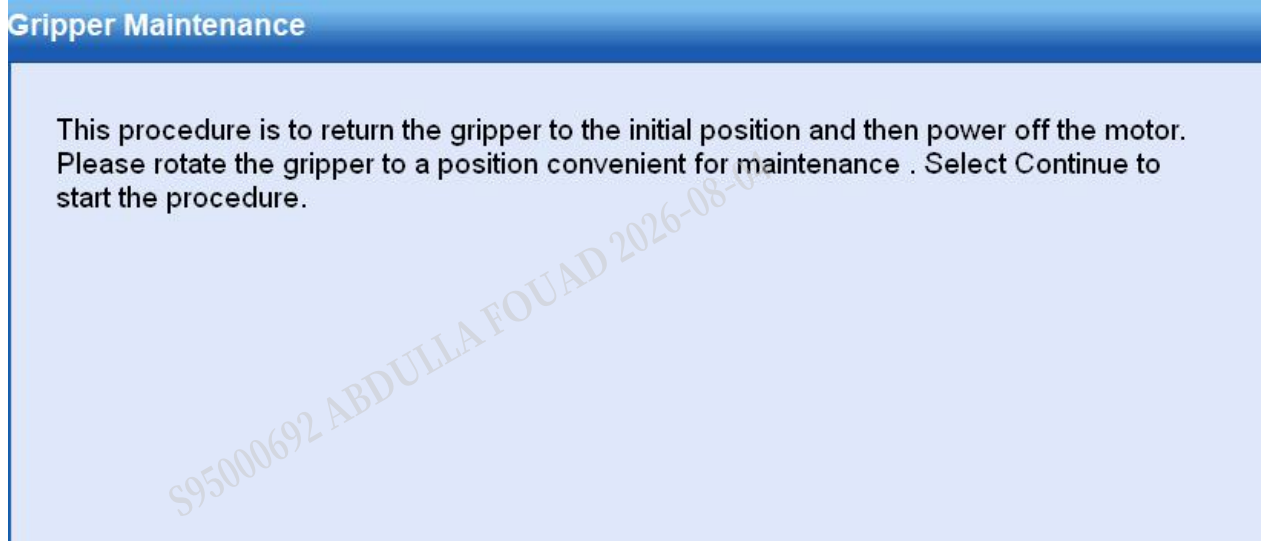
Supplies Required: Dust-free gauze or cotton swab, and ethanol

Precautions: Wear gloves before operation to prevent biohazards

Analyzer Status: Idle or stop

Procedure:

- a. Select **Maintenance > Maintenance Command > Gripper Maintenance**, and click **Continue**.



- b. Use the screwdriver to loosen the screws on the gripper shield and remove the covers
- c. Use the dust-free gauze moistened with ethanol to clean the fingers until they are free of any dust, dirt, or rust and shows metallic luster.
- d. After the mechanical reset is completed, jump to the gripper spring replacement interface, and complete the spring replacement of the first/second grippers according to the text prompt; loosen the fastening screw of the Z-axis housing with a Phillips screwdriver, and remove the Z-axis housing.
- e. When replacing the gripper spring, turn the gripper cam with tweezers to close the fingers. Remove the spring from the post with tweezers, and pay attention not to drop the spring or any other parts into the instruments (the reaction carousel hole or the covers).
- f. Install a new finger clamping spring by hanging the spring hook to the groove of the spring guide post.
- g. Click **Continue** to enter the gripper finger cleaning interface, and follow the text prompts to clean the fingers.
- h. Use the dust-free gauze moistened with ethanol to clean the fingers until they are free of any dust, dirt, or rust and shows metallic luster.
- i. Restore the gripper covers.
- j. Select **Continue** to reset all mechanical parts of the whole unit.
- k. Click **Done** to end the maintenance process.

Confirmation Method: The fingers should be free of any dust, dirt, or rust and show metal brilliance.

Maintenance Duration: 2min

2.2.14 Replace the Spring of Gripper

Maintenance Object: Gripper spring

Timing: One year

Maintenance Cause: A spring may fall off or be broken after the instrument has been used for a long time, in which case the gripper function will be impaired.

Maintenance Item: Replace the gripper spring

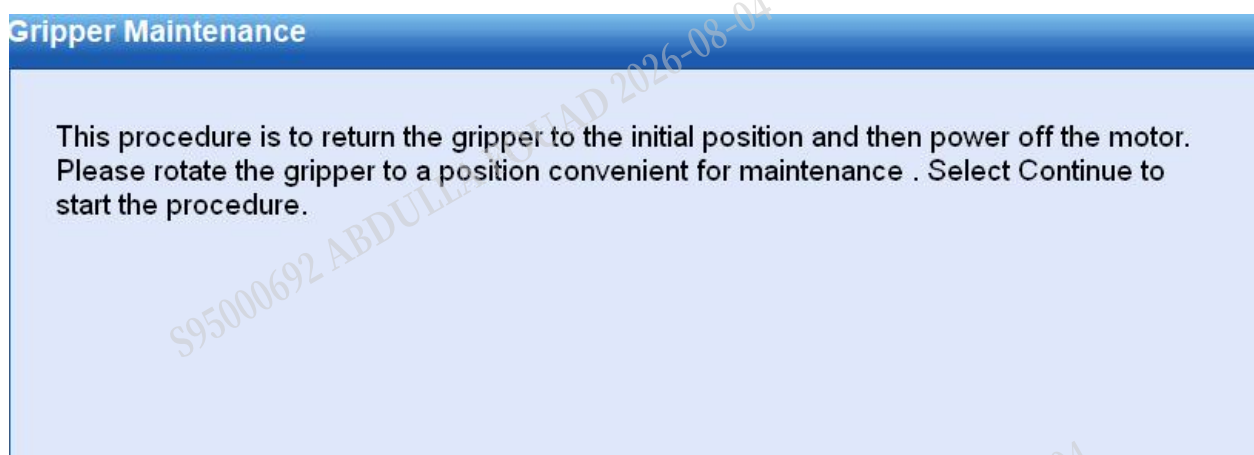
Supplies Required: New gripper spring (in the maintenance kit), dispersion aspirating probe unclogging tool

Precautions: Wear gloves before operation to prevent biohazards

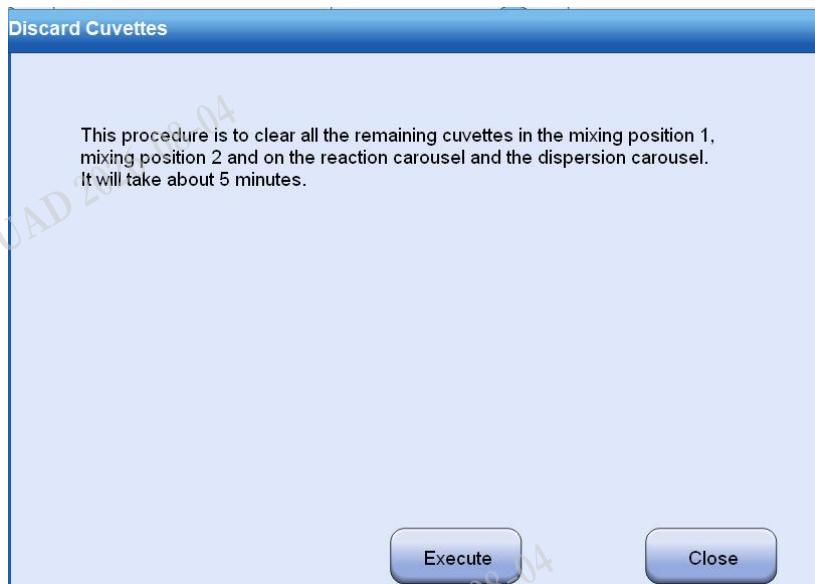
Analyzer Status: Idle or stop

Procedure:

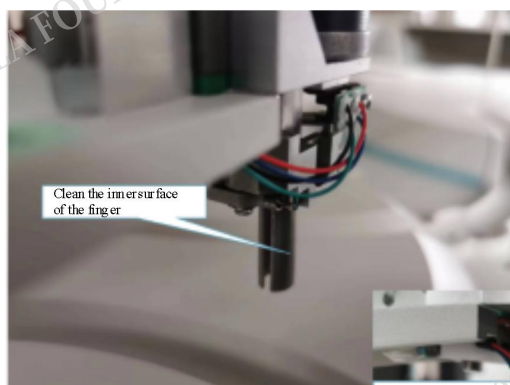
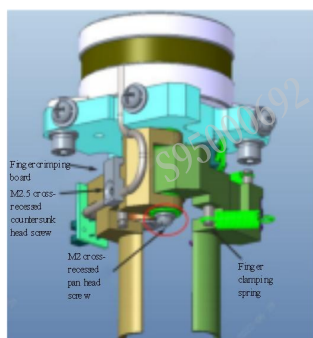
- a. Select **Maintenance > Maintenance Command > Gripper Maintenance**, and click **Continue**.



- b. Use the screwdriver to loosen the screws on the gripper shield and remove the covers
- c. When replacing the gripper spring, turn the gripper cam with tweezers to close the fingers. Using the ring end of the dispersion aspirating probe unclogging tool to hook the end of the spring and remove the spring from the post. Be careful not to drop the finger spring or any other finger components into the instrument (e.g., into the reaction carousel holes or other instruments).
- d. Hang a new gripper spring to the groove of the post.
- e. Restore the gripper covers.
- f. Select **Done** to exit the maintenance procedure.
- g. Select **Maintenance Command > Discard Cuvettes** and check the opening/closing function of the gripper fingers.



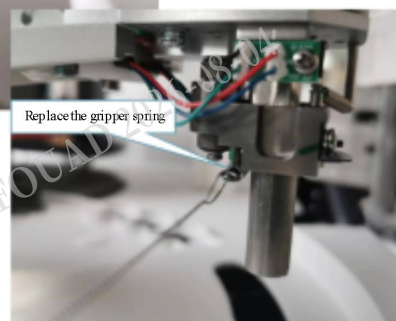
Confirmation Method: The gripper can normally grip the cuvettes.
Maintenance Duration: 10min



Unclogging tool for dispersion aspirating probes



Finger spring



2.2.15 Align the Mechanical Position

Maintenance object: Positions of sampling probe, gripper, sample rack delivery, and dispersion carousel unit

Maintenance Timing: Six-month

Maintenance Cause: After long-term use, the relative position of mechanical moving parts may deviate, and they need to be corrected again.

Maintenance Item: Confirm the position of sampling probe/gripper/sample rack delivery/dispersion unit

Supplies Required: Alignment tooling

Precautions:

1. To protect against biological hazards, wear disposable rubber gloves at all times during the entire alignment process. Avoid direct skin contact with any internal components of the instrument.
2. If any injury like puncture wound, incised wound or bruise occurs, clean the hands and the injured sites,

use proper skin disinfectant, and seek medical treatment if necessary. Record the injury causes and keep the complete medical record.

3. After alignment, remove alignment fixtures from the instrument to avoid collision.
4. When the axis of the alignment fixtures (pseudo probe or positioning axis) align with the center of the fixture hole (pseudo cuvette), let them align with each other automatically and do not hold them up with your hands.
5. When checking if the alignment fixture axis aligns with the hole, observe them from different angles with a difference of at least 90°.
6. When operating with the sampling probe, dispensing probe, and aspirating probe, please wear disposable gloves and handle them gently to avoid the deformation.

Instrument Status: In the Idle or Stop status

Procedure:

- a. Refer to the Alignment Guide section of the Service Manual

Maintenance Duration: 30min

2.2.16 Execute Daily Clean

Maintenance Object: Sampling probes and dispersion aspirating probes

Maintenance Timing: Six-month

Maintenance Cause: Keep good performance of probes, avoid probe clogging and reduce influence of the probe carryover on the test result.

Maintenance Item: Check if aspirating probes are clogged and clean the sampling probes and dispersion aspirating probes

Supplies Required: About 3.6 ml of CD80 concentrated wash solution is consumed.

Precautions: Before operation, wear gloves to prevent biological risks; During operation, keep a safe distance from moving parts such as the sampling probe to avoid collision.

Instrument Status: Idle

Procedure:

- 1) Select **Utility > Maintenance > Maintenance > Maintenance Command**.
- 2) Select **Daily Clean**.
- 3) Add sufficient concentrated wash solution into the 60 ml reagent bottle, and place it in the special wash position D3.
- 4) Click **Execute**;
- 5) To stop test, click **Stop**.
- 6) After the daily cleaning is completed successfully, select **Exit** to close the window..
- 7) Select **Utility > Maintenance > Maintenance**, and then select **Scheduled Maintenance > Daily**.

- 8) Tick the checkbox to the right of **Execute Daily Clean**.
- 9) Click **OK** to refresh the current date as the performance date.
- 10) Click **Log**, which record errors that occurred and other necessary information for the procedure.
- 11) Click **OK** to save the log.

Note: Log path:

D:\Mindray\CL1000i\OperationSoft\Instrument_1\MaintenRecordLog\DailyCleanClogCheck.csv

Confirmation Method: After maintenance, repeat the above procedure to see if it can be completed normally.

Maintenance Duration: 20min

2.2.17 Run System Check

Maintenance Object: Confirm the performance of the whole unit

Maintenance Timing: Six-month

Maintenance Cause: Ensure the performance meets the requirement.

Maintenance Item: Background test, wash buffer check, and probe dispensing performance test

Supplies Required: System check solution (free enzyme D)

Precautions: Wear gloves before operation to prevent biohazards

Instrument Status: Idle

Procedure:

- a. Select **Utility > Maintenance > Diagnosis > System Diagnosis**.
- b. Just select **Background Test, Wash Buffer Check, and Dispensing Performance Test**.

System Check

System check is performed for weekly maintenance, system diagnosis and performance test.
This procedure will take about 45 minutes.
Dispense the following solution on the rack according to the selected test panel.
Sample tube 1: 1ml system check solution 1 placed in position 1 of the rack (used for the precision test of the probe)
Sample tube 2: 2ml system check solution 2 placed in position 2 of the rack (used for cleaning test).

Lane

☒ Background Test

☒ Wash Buffer Check

☒ Probe Performance Test

☐ Wash Test

☐ Cleaning Test

☒ None/ All

- c. Prepare a sample rack and load 1 ml of system check solution in position 1# of the rack; Load the sample rack to the lane 1 position; click **Execute** and wait for results.

d. The interface after test is as follows:

系统检测报告							
完成时间:2015-08-24.19:31							
	Limits	Value	Status				
洗液污染	-500 ~ 500	-646	Fail				
检测项目	AVE	Limits	Status	SD	CV(%)	Limits	Status
本底检测	3357	3000 ~ 6000	Pass	83	/	0 ~ 200	Pass
清洗液检测	476	/	/	/	/	/	/
加样性能检测	165859	/	/	/	0.94	0.00 ~ 2.00	Pass
名称	序号	光计数					
本底检测	9	3408					
本底检测	10	3460					
加样性能检测	1	166767					
加样性能检测	2	164411					

Confirmation Method: All the selected test items can pass the test.

Maintenance Duration: 40min

Note: Log path:

D:\Mindray\CL1000i\OperationSoft\WorkStationLOG\测试日期\FlowSrvPerf

2.2.18 Execute Effect Check

Maintenance Object: Sampling probe, dispersion assembly, and PMT module

Maintenance Timing: Six-month

Maintenance Cause: Effect check is used to avoid crystallization in probe washing tubes, dispersion dispensing tubes and substrate dispensing tubes and monitor the liquid contamination and the working status of the photometric system.

Maintenance Item: The liquid contamination and the working status of the photometric system are monitored.

Supplies Required: None

Precautions: Wear gloves before operation to prevent biohazards

Instrument Status: Idle

Procedure:

- 1) Select **Utility > Maintenance > Maintenance > Maintenance Command**.
- 2) Select **Effect Check**.
- 3) Select **Utility > System Setup > Instrument > Auto Maintenance Settings** to set the interval of

effect check as 4 hours by default.

- 4) Click **Execute**. To pause the test, click **Stop**;
- 5) After the procedure, the photometric count of each cuvette position will be displayed on the interface;
- 6) Click **Exit** to close the window;
- 7) Select **Utility > Maintenance > Maintenance**, and then select **Scheduled Maintenance > Daily**.
- 8) Tick the checkbox to the right of **Execute Effect Check**.
- 9) Click **OK** to refresh the current date as the performance date.
- 10) Click **Log**, which records errors that occurred and other necessary information for the procedure.
- 11) Click **OK** to save the log.

Confirmation Method: Effect check can be completed normally.

Maintenance Duration: 15min

Note: Log path:

D:\Mindray\CL1000i\OperationSoft\Instrument_1\MaintenRecordLog\effectTestLog.csv

2.2.19 Statistics of Component Service Life

Maintenance Object: Each component with its service life

Timing: One year

Maintenance Cause: After the component is replaced, the service time needs to be cleared.

Maintenance Item: Check the service life of each component and reset the service life of the new device.

Supplies Required: None

Precautions: Do not clear the service time of a component that has not been replaced

Instrument Status: Idle

Procedure:

- a. Select **Utility > Maintenance > Maintenance Command > Statistics of Component Service Life**.

Maintenance

Scheduled Maintenance	Maintenance Command	Engineer
	Home	Discard Cuvettes
	Clean Probe Exterior	Fluidic Prime
	Clean Probe Interior	Effect Check
	Clean/Replace Probe	Daily Clean
	Syringe Maintenance	Sensor Calibration
	Clean Vortexer	Gripper Maintenance
	Clean Dispersion Probes/Tubes	Statistics of Component Service Life
		Photometer HV correction

Close

Statistics of Component Service Life

Name	Record the start time	Accumulative operation
Syringe	2025-12-2 14:22:52	5998time(s)
Probe Interior Wash Syringe	2025-12-1 19:24:07	6428time(s)
Probe Exterior Wash Syringe	2025-12-1 19:24:07	7986time(s)
Dispersion syringe	2025-12-1 18:08:32	5484time(s)
Substrate syringe	2025-12-1 11:39:10	4014time(s)
Gripper	2025-11-28 14:17:57	20939time(s)
Waste vacuum pump	2025-11-28 12:59:12	101Hour
Inlet Pump	2025-11-28 13:07:41	3Hour
Dispersion Peristaltic Pump	2025-12-1 15:40:34	12Hour
Photometer	2025-12-1 18:22:51	88Hour

Clear Close

- Select the maintenance item and click the **Clear**
- Components that are replaced during proactive maintenance are typically:

- d. Dispersion peristaltic pump tube and gripper spring
- e. After updating, click **Exit** in the lower right corner.

Confirmation Method: /

Maintenance Duration: 2min

2.2.20 Check the substrate leakage box

Maintenance Object: substrate leakage box

Timing: One year

Maintenance Cause: When substrate tubing is cleaned or the substrate bottle is removed, small amount of liquid will leak into the substrate holder and flow into the substrate leakage box, so the substrate leakage box must be cleaned regularly.

Maintenance Item: Check and clean substrate leakage box

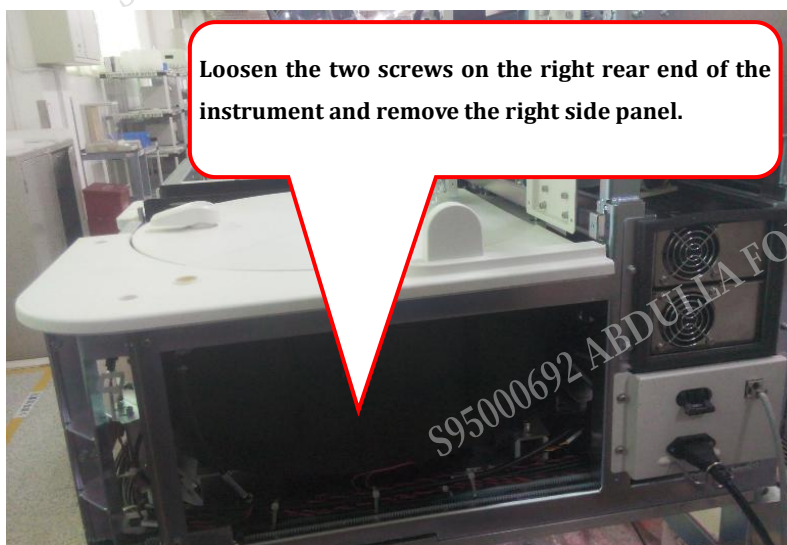
Supplies Required: Brush, clean gauze, and DI water

Precautions: Wear gloves before operation to prevent biohazards

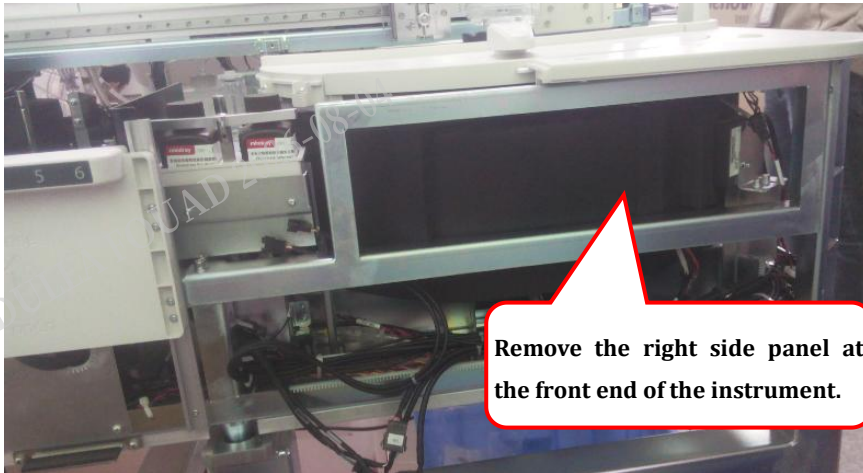
Instrument Status: The instrument must be **powered off**

Procedure:

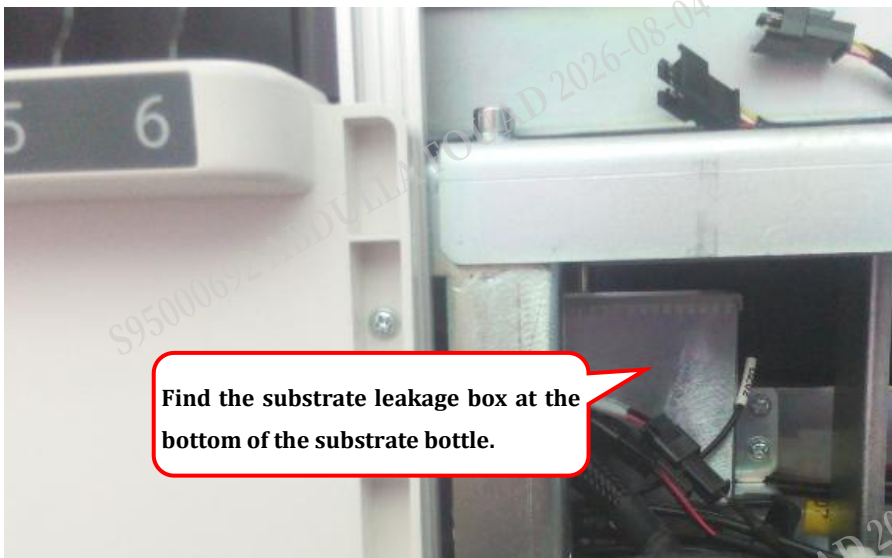
- a. Loosen the two screws on the right rear side of the instrument and remove the side panel of the instrument.



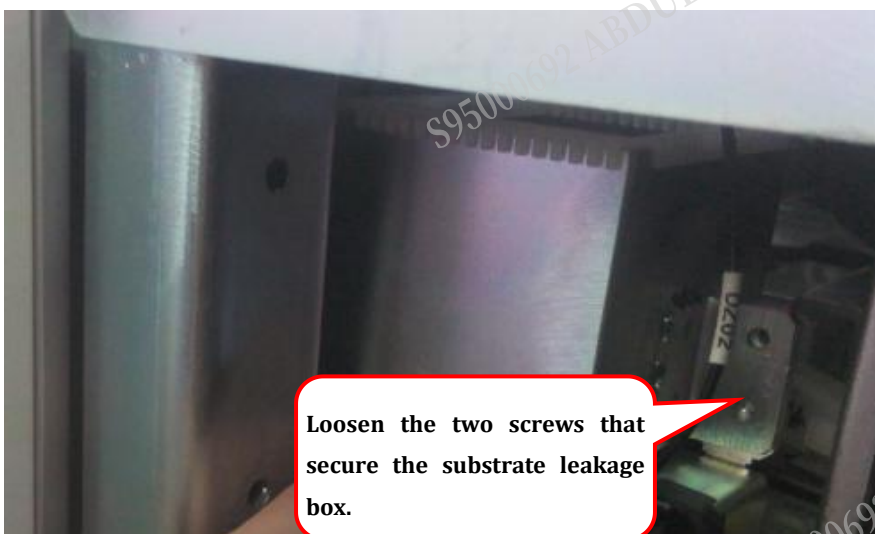
- b. Remove the right side panel at the front end of the instrument.



- c. Find the substrate leakage box at the bottom of the substrate bottle.



- d. Loosen the two screws that secure the substrate leakage box and carefully remove the substrate leakage box.



Note: Be careful not to pull the substrate tubes!

- e. Check the substrate leakage box to see whether there is any fluid. If so, pour it off.
- f. Rinse the substrate leakage box with tap water and then wash it with a hair brush.
- g. After rinsing, dry the substrate leakage box.

h. Install the substrate leakage box in accordance with the reverse operation of the above steps 1-4.

Confirmation Method: Visually inspect that substrate leakage box is clean and dry.

Maintenance Duration: 25min

2.2.21 Clean the Cooling Fans

Maintenance Object: Refrigerating fans 1 and 2, and cooling fan of whole unit

Timing: One year

Maintenance Cause: After long-term operation, there will be dust on the fan, which will affect the heat dissipation effect.

Maintenance Item: Clean the dust on the cooling fan

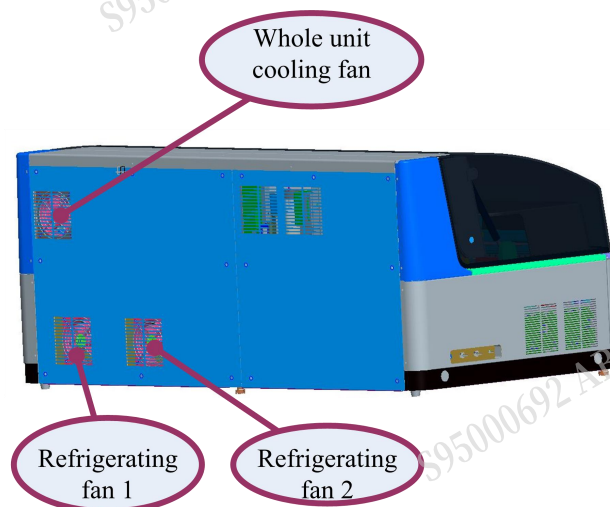
Supplies Required: Brush, clean gauze, and DI water

Precautions: Wear gloves before operation to prevent biohazards

Instrument Status: The instrument must be **powered off**

Procedure:

- a. Shut down (confirm that the main power is off)
- b. Remove the shield cover and rear cover.
- c. Find fans for refrigeration 1 and 2, and whole unit cooling fan.



- d. Remove dust from the fan surface (Remove the fan if necessary).
- e. At the same time, clean the fan air outlet and the dust on the blinds.
- f. Restore the fans.
- g. Power on the whole unit, confirm that the fan works normally and restore the shield cover.

Confirmation Method: Enter the operating software, query the working status of the fan: Select **Utility > Maintenance > Alignment > Temperature Control > Fan Status Query**, and click **Search**.

Fan Status Query

Cooling Fan of Whole Unit
Cooling Fan of Reagent Chamber
Refrigeration Duct Fan 1(left)
Refrigeration Duct Fan 2(right)
Refrigeration Hot End Fan

Search Exit

Maintenance Duration: 10min

2.2.22 Check the vacuum waste pump

Maintenance Object: Vacuum waste pump

Timing: One year

Maintenance Cause: After long-term use, a small amount of liquid in the pipeline may accumulate or crystallize inside the pump head, affecting the pump performance.

Maintenance Item: Determine whether the air pump can normally establish vacuum

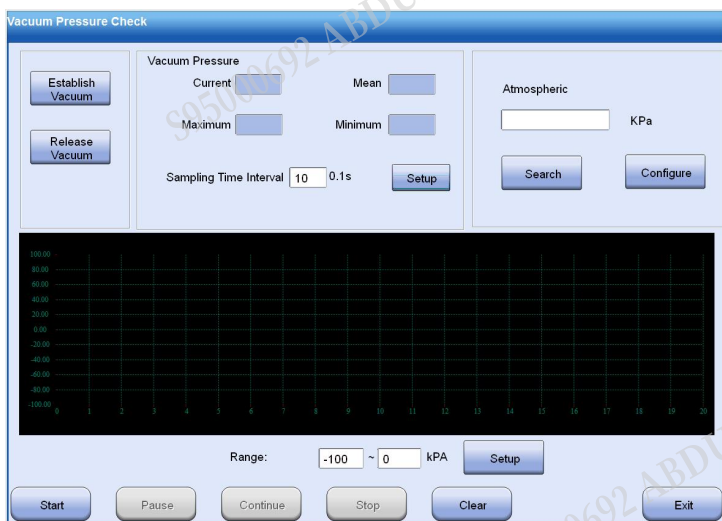
Supplies Required: None

Precautions: Only engineers can perform maintenance. Wear gloves before operation to prevent biological risk

Instrument Status: Idle

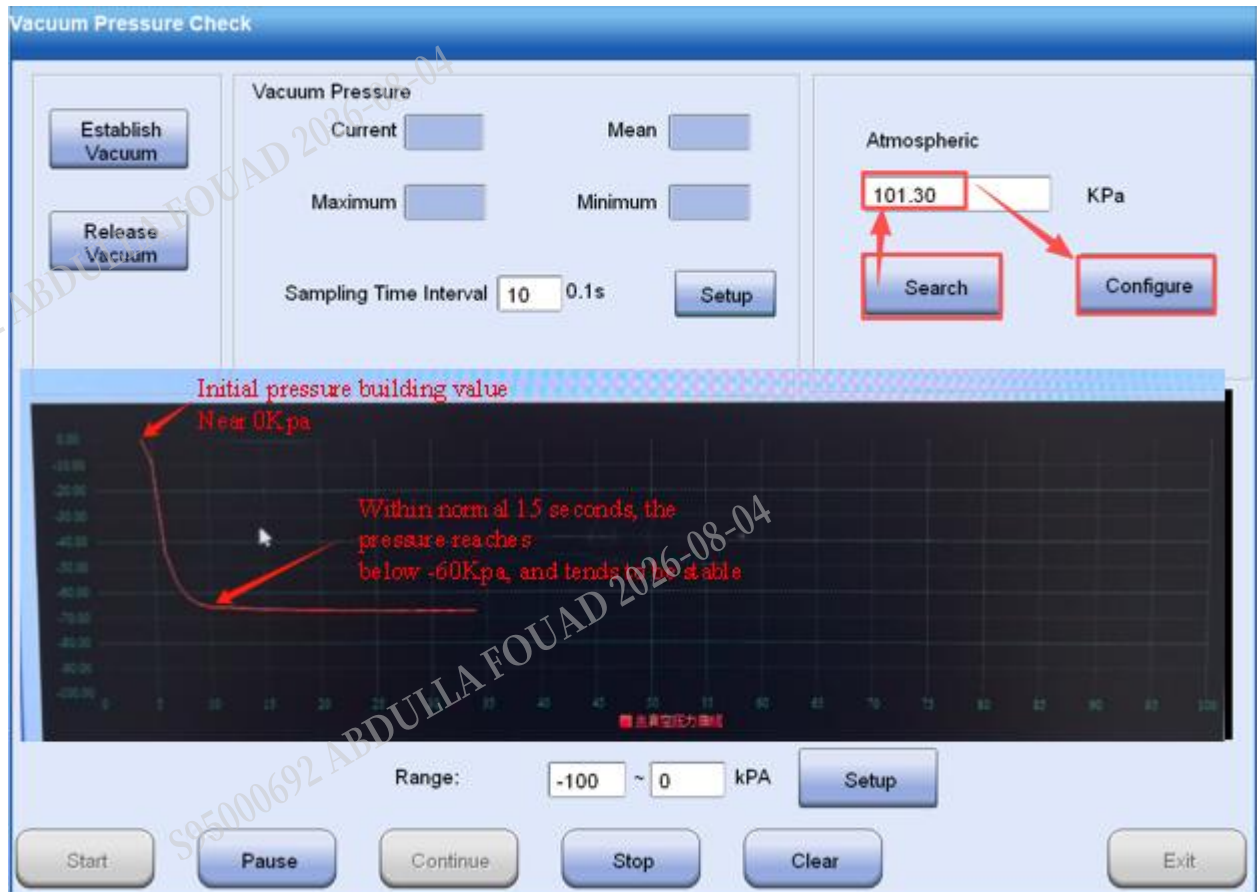
Procedure:

- Operate the software and enter the Fluidics Alignment screen.
- Click Vacuum Pressure Detection to enter the pressure detection screen.
- Confirm that Local Atmospheric Pressure is set correctly.
- Select Create Vacuum (configure the external air pump instrument, and select the external air pump at the same time), and click Start. The air pump starts. Click Start on the pressure collection screen to start collecting pressure.



- The system automatically completes the pressure detection and prompts whether the detection is successful. After completion, click the Pressure Collection Stop button and the Air Pump Stop button.
- If the test fails, the air pump module may need maintenance.
- Exit the screen and the system is recovered.

Confirmation Method: Whether the vacuum is established successfully



Maintenance Duration: 5min

2.2.23 Check the refrigeration current

Maintenance Object: Refrigeration module

Timing: One year

Maintenance Cause: The common failure mode of refrigeration module is that the condensate seeps into the interior of refrigeration module, resulting in short circuit of Peltiers and abnormal power supply current. Through the engineer's active inspection, it is expected that the abnormal refrigeration module will be identified and handled in advance before the problem occurs.

Maintenance Item: Refrigeration Module

Supplies Required: None

Precautions: Only engineers can perform maintenance. Wear gloves before operation to prevent biological risk

Instrument Status: Idle

Procedure:

- Select Utility > Status > Power Supply.

	Chemistries	System Setup	Maintenance	Status	Update parameters
Program	Status Summary	Temperature	Power	Hydro	Smart Modules
	Name	Measured Value	Valid Range		
Result	Main Board Other -12V Analog Voltage	0.00	-12.90 V - -11.10 V		
	Main Board 24V Voltage	0.00	22.30 V - 25.80 V		
	Main Board 12V Voltage	0.00	11.10 V - 12.90 V		
Reagent	Main Board Other +12V Analog Voltage	0.00	11.10 V - 12.90 V		
	Photometer -12V Analog Voltage	0.00	-12.90 V - -11.10 V		
	Main Board 5V Voltage	0.00	4.50 V - 5.45 V		
QC	Main Board 3.3V Voltage	0.00	3.12 V - 3.49 V		
	Photometer +12V Analog Voltage	0.00	11.10 V - 12.90 V		
	Reagent Refrigeration Board Fan +24V	N/A	22.30 V - 25.80 V		
Utility	Reagent Refrigeration Board B12V	N/A	11.10 V - 12.90 V		
	Reagent Refrigeration Board F12V	N/A	11.10 V - 12.90 V		
	Reagent Refrigeration Board Digital +5V	N/A	4.50 V - 5.45 V		
Alarm	Reagent Refrigeration Board Digital +3.3V	N/A	3.12 V - 3.49 V		
	Radiator 1 Current	N/A	2.0 A - 6.5 A		
	Radiator 2 Current	N/A	2.0 A - 6.5 A		
	Radiator 3 Current	N/A	2.0 A - 6.5 A		
	Radiator 4 Current	N/A	2.0 A - 6.5 A		

- Check whether the current of the four Peltiers is between 2.0A and 6.5A
- If the current is out of range, change the refrigeration module. For details, please refer to the Service Manual

Confirmation Method: Check whether the current of the four Peltiers is between 2.0A and 6.5A

Maintenance Duration: 2min

2.2.24 Verification After Maintenance:

- After maintenance, there is no fault in operation.
- QC meets the requirements.
- Specimen meets the requirements.
- The reserved samples are compared and meet the requirements.

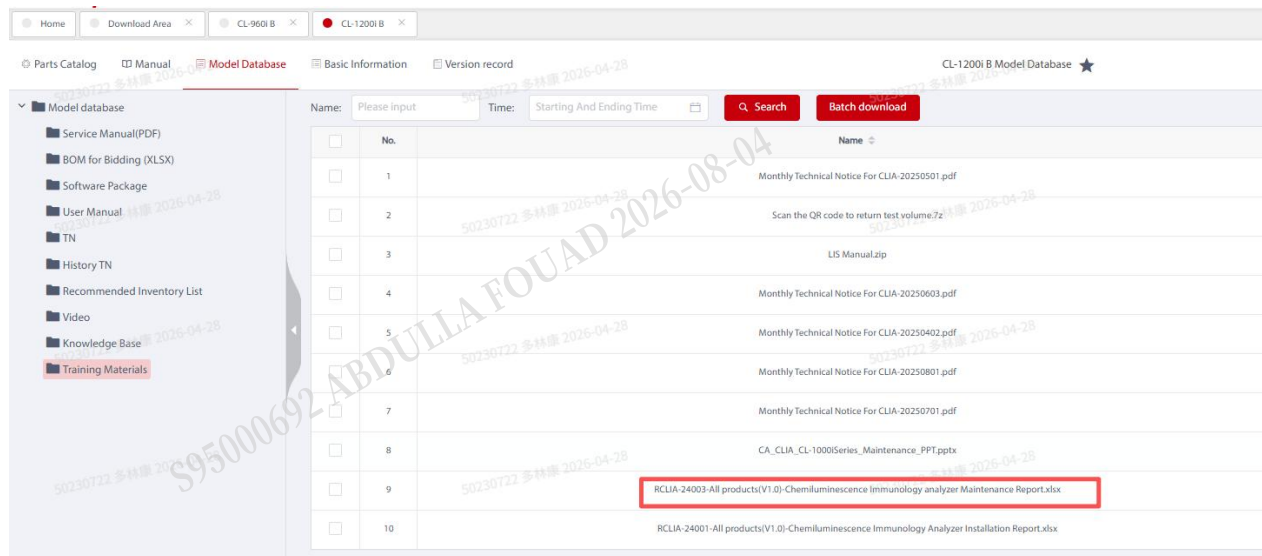
Chapter 3 Maintenance Report

3.1 Output the Maintenance Report

3.1.1 Print a blank maintenance report in advance

Download path: Maintenance report for the corresponding model on the TIS platform. Fill in the corresponding report according to the performed maintenance and complete it.

File name: *RCLIA-24003-All products(V1.0)-Chemiluminescence Immunology analyzer Maintenance Report.xlsx*



3.2 Process and Record the Maintenance Work Order

3.2.1 Requirements

Fill in the header and maintenance item of the report form truthfully, in which the customer's signature in the footer needs to be **signed by** the person in charge or the director of the department (**it is better to note the contact phone number**), and **take a photo for retention**.

3.2.2 Record Upload Requirements

After Maintenance:

The engineer can change the work order status to on-site service completion in the MSP system, and upload the original signature of the on-site service work order and the original signature of the maintenance report. In addition, the engineer needs to upload the photos of the maintenance parts.